Investing in AI for Investing: Lessons from Autonomous Driving for Investment Firms

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Abstract

We build on the successful SAE J3016 standard for autonomous driving to introduce a similar taxonomy (L0-L5) for autonomous investing systems. The framework, which ranges from no automation (Level 0) to full automation (Level 5), delineates system capabilities, human oversight requirements, and operational boundaries for AI-driven investment processes. We present three industry case studies across analysis for a fundamental equities manager for sleep apnea devices, evaluations of decentralized finance protocols, and manager oversight for a sovereign fund manager. As in the field of autonomous driving, we hope the framework can enable clear communication between investors, policymakers, and the public while helping to set clear expectations for system capabilities and human responsibilities. This is especially needed because while almost three-quarters of asset managers expect generative AI to transform their industry within five years, only one in six have comprehensive AI strategies, highlighting the need for standardized approaches to AI implementation.

Keywords: Artificial intelligence, investment management, asset management, DeFi, autonomous systems, financial technology, risk management, regulatory framework

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1 Introduction

There is a striking gap in the adoption of AI in the investment management industry. Recent industry surveys show that 72% asset managers expect generative AI to create transformative changes over the next three to five years.[1] At the same time, only 16% asset managers have developed comprehensive strategies to implement these technologies. The disparity is more pronounced considering that in other industries like manufacturing, healthcare, transportation, and communications, AI has already upended long-established workflows—and yet in financial services, firms are still grappling with how to start pilot projects. This divide represents both a significant opportunity and an urgent challenge: Investment firms that can bridge the gap between high expectations and concrete implementation will redefine how investment decisions are made, portfolios are constructed, and risks are managed.

The challenge is compounded by the opacity of many modern AI systems. The "black-box" models arrive fully trained, with internal decision pathways that are not clear even to their developers. In the investment management industry—which has fiduciary responsibilities and faces regulatory oversight and compliance—there are difficult questions: How do you validate a trading signal that you cannot dissect? Who bears responsibility when an AI-generated recommendation goes awry? And how do you satisfy auditors and regulators demanding clear audit trails? Overcoming this 'explainability' hurdle is essential not just for compliance but for building the trust and governance frameworks that will enable AI to deliver on its promise in asset management.

This paper addresses these challenges by proposing a systematic framework for classifying AI autonomy in investment management. Drawing inspiration from the highly successful SAE J3016 standard for autonomous driving, we introduce a six-level taxonomy that provides clear definitions of system capabilities, human responsibilities, and operational boundaries for AI-driven investment systems. The autonomous driving industry faced similar challenges in the early stages of AI development. Multiple companies were developing systems with varying capabilities, but without common terminology or clear standards for comparing different approaches. The introduction of the SAE J3016 standard created a shared language that allowed more effective communication among engineers, policymakers, and the public while establishing clear expectations for system performance and human oversight.

Investment management shares many characteristics with autonomous driving that make this analogy particularly relevant. Both domains involve complex, dynamic environments that require continuous assessment of evolving conditions. Both require sophisticated decision-making under uncertainty, with significant consequences for poor choices. Both investment management and autonomous driving involve intricate risk evaluation processes and the potential for meaningful human-AI collaboration. Most importantly, both require clear delineation of responsibilities between human operators and automated systems.

Our proposed framework aims to provide the investment industry with the same benefits that the SAE standard has delivered for autonomous driving: improved clarity in discussions about AI capa-

[1] See https://www.bcg.com/press/6may2024-asset-managers-leverage-ai-opportunity-and-drive-future-growth.

bilities, systematic guidance for technology development, and transparent frameworks for regulatory oversight and risk management.

2 Literature Review and Theoretical Foundation

We first outline the main framework of the SAE J3016 standard for autonomous driving, which we will apply to investment management. We discuss the state of AI adoption in the investment industry and the need for standardization of AI terms.

2.1 The SAE J3016 Standard for Autonomous Driving

The Society of Automotive Engineers (SAE) J3016 standard, "Levels of Driving Automation," has emerged as the definitive global framework for classifying automated driving systems. First published in 2014 and regularly updated to reflect technological advances, this standard has achieved remarkable adoption across industry, academia, and regulatory bodies worldwide.

The success of the SAE J3016 standard stems from its clear definition of several core concepts that allow precise communication about the capabilities and limitations of the system. The Dynamic Driving Task encompasses all real-time operational and tactical functions required to operate a vehicle, including steering, acceleration, deceleration, and obstacle avoidance. The Automated Driving System refers to the combined hardware and software that performs driving functions. Object and Event Detection and Response represents the critical capability to monitor the environment, identify relevant objects and events, and execute appropriate responses.

Perhaps most importantly, the standard defines the Operational Design Domain—the specific conditions under which a given system is designed to function—and establishes clear human roles and fallback responsibilities at each level of automation. This structured approach to human-system interaction has proven essential for safety, regulatory compliance, and public acceptance of autonomous driving technologies.

The six levels of the SAE standard progress from Level 0 (no automation) through Level 5 (full automation), with each level clearly defining system capabilities and human responsibilities. At Levels 0 to 2, humans remain responsible for monitoring the driving environment and maintaining readiness to take control. Level 3 represents a critical transition where the system monitors the environment but humans must respond to requests for intervention. Levels 4 and 5 eliminate the need for human monitoring within defined operational domains (Level 4) or unlimited domains (Level 5).

2.2 AI Applications in Investment Management

The application of artificial intelligence in investment management has evolved rapidly over the past decade, encompassing a broad range of functions from basic data processing to sophisticated portfolio optimization. Current AI applications span multiple categories of investment tasks, each with different

requirements for human oversight and different implications for risk management.

Data acquisition and processing represent the most established applications of AI in finance. These systems automate the collection and structuring of financial documents, earnings call transcripts, news articles, and regulatory filings. Natural language processing techniques enable extraction of key metrics and trends from unstructured documents, significantly reducing the manual effort required for fundamental analysis.

Quantitative analysis and valuation support have seen increasing AI adoption, with systems capable of calculating financial metrics, identifying anomalies in company fundamentals, and contributing to valuation frameworks such as discounted cash flow models. These applications can reduce the computational burden on human analysts while enabling analysis of larger datasets than traditional manual approaches.

Qualitative factor assessment represents a more sophisticated application area where AI systems analyze narrative sections of financial reports, assess management commentary tone and sentiment, and detect shifts in executive communication that may signal business changes. Advanced systems now incorporate audio and video analysis of earnings calls and management presentations.

Signal generation and hypothesis formation represent emerging frontiers where AI systems can detect patterns or correlations in company fundamentals that suggest investment opportunities. These systems highlight companies exhibiting characteristics historically associated with superior returns, based on specific investment philosophies or quantitative models.

Risk management applications have gained particular attention following several high-profile algorithmic trading failures. AI systems now identify unusual financial behavior, evaluate downside scenarios, flag emerging risks within portfolios, and highlight important information in regulatory filings that might otherwise be overlooked.

Finally portfolio construction and monitoring represent some of the most advanced current applications, where AI systems rebalance positions based on real-time assessments of company performance or valuation changes. These systems can alert managers to deviations from strategic asset allocations and signal when company fundamentals have diverged from expectations.

2.3 Need for Standardization

The current landscape of AI applications in investment management lacks the standardized terminology and classification frameworks that have proven essential in other technology-intensive industries. This absence of common standards creates several significant challenges for firms seeking to implement AI technologies effectively.

Communication barriers emerge when different firms, technology vendors, and regulatory bodies use different terminology to describe similar AI capabilities. These barriers impede effective collaboration, technology evaluation, and regulatory oversight. Without clear definitions of system capabilities and limitations, firms struggle to make informed decisions about technology adoption and integration.

Risk management becomes significantly more complex when AI systems operate without clear

classification of their capabilities and operational boundaries. Investment firms have fiduciary responsibilities that require understanding of how investment decisions are made and who bears responsibility for outcomes. The "black box" nature of many AI systems compounds these challenges by making it difficult to assess system reliability and identify potential failure modes.

Regulatory compliance represents another area where standardization becomes critical. Financial regulators increasingly focus on AI governance, algorithmic accountability, and systemic risk management. Clear frameworks for classifying AI capabilities and defining human oversight responsibilities will likely become essential for demonstrating regulatory compliance as oversight frameworks evolve.

Technology development and evaluation benefit significantly from standardized frameworks that enable systematic comparison of different AI approaches. Just as the SAE standard has facilitated progress in autonomous driving by providing clear targets for technology development, a similar framework for investment management could accelerate innovation while ensuring appropriate risk management.

3 The L0–L5 Framework for Autonomous Investing

Our proposed framework adapts the core principles of the SAE J3016 standard to the domain of investment management while preserving the clarity and systematic structure that have made the driving automation standard so successful.

3.1 Framework Principles and Definitions

The framework defines six levels of autonomy (L0–L5) with clear delineations of system capabilities, human responsibilities, and operational boundaries at each level.

- **Dynamic Investment Task (DIT)**: We define the Dynamic Investment Task as the overall process being automated, encompassing all real-time operational and tactical functions required for investment management. This includes fundamental analysis, valuation modeling, portfolio construction, risk assessment, trade execution, and performance monitoring. The DIT represents the complete scope of activities that an autonomous investment system might perform.

- **Autonomous Investment System (AIS)**: The Autonomous Investment System comprises the hardware and software components that together perform investment-related tasks. This includes data processing infrastructure, analytical models, decision-making algorithms, and execution systems. The AIS capabilities determine the level of autonomy that can be achieved within specific operational domains.

- **Information and Event Detection and Response (IEDR)**: A critical subset of the Dynamic Investment Task that involves monitoring financial markets and companies, detecting and classifying relevant information and events, and executing appropriate investment responses. The

capability of the AIS to perform IEDR effectively distinguishes different levels of autonomy and determines the system's ability to operate independently.

- **Operational Design Domain (ODD)**: The specific conditions under which a given autonomous investment system is designed to function. This includes asset classes, geographic markets, investment strategies, market conditions, and risk parameters. The ODD clearly defines the boundaries within which the system can operate autonomously and identifies conditions that require human intervention.

- **Human Role and Fallback**: Clear delineation of human responsibilities at each level of autonomy, including monitoring requirements, intervention protocols, and fallback procedures when systems encounter situations outside their operational design domains or experience failures.

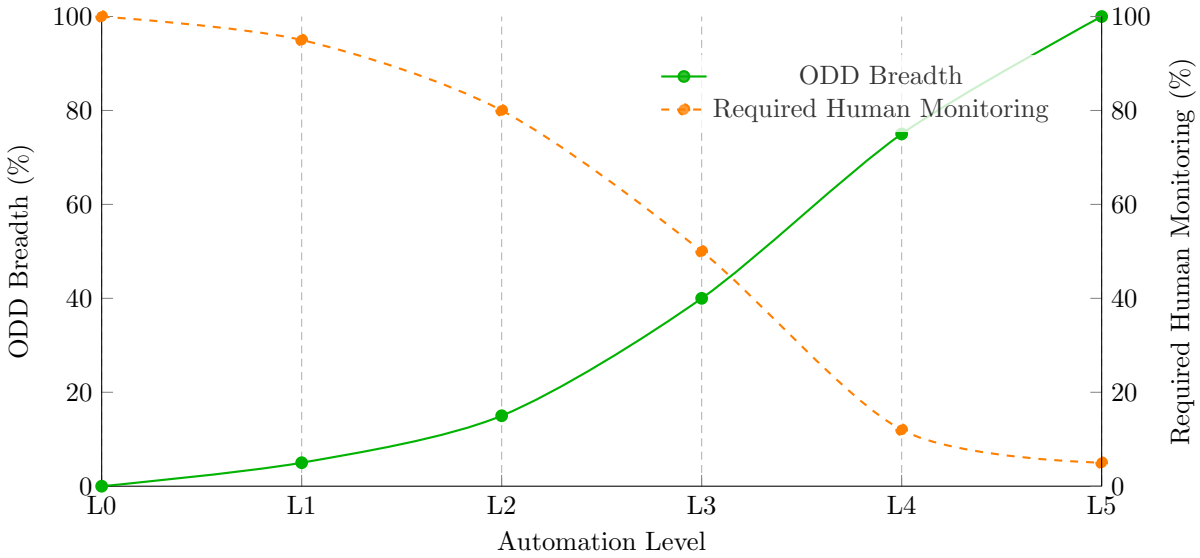

Figure 1: As ODD expands, required human monitoring decreases.

3.2 Human-AI Interaction Models

The framework incorporates three distinct models of human-AI interaction that correspond to different levels of autonomy and have important implications for risk management and regulatory compliance.

- **Human-in-the-Loop (HITL)**: In this model, humans maintain active involvement in all critical decisions while AI systems provide analysis, recommendations, and support. The human retains final authority over investment decisions and continuously validates AI outputs. This model applies primarily to L1 and L2 systems where AI capabilities are limited to assistance and partial automation.

- **Human-on-the-Loop (HOTL)**: This model involves AI systems making routine decisions within defined parameters while humans monitor system performance and intervene when necessary. Humans set strategic direction, define operational boundaries, and respond to alerts when

systems encounter unusual situations. This model characterizes L3 systems where conditional automation operates under human supervision.

- **Human-out-of-the-Loop (HOOTL)**: In this model, AI systems operate autonomously within their defined operational domains without routine human intervention. Humans establish high-level objectives, risk parameters, and ethical guidelines but do not participate in tactical decision-making. This model applies to L4 and L5 systems where high or full automation eliminates the need for continuous human oversight.

3.3 Detailed Level Descriptions

Analogous to SAE J3016 for autonomous driving, we define the following levels L0 to L5 for employing AI in investment management:

- **Level 0:** ***No Investment Automation.*** Investment professionals perform the entire Dynamic Investment Task manually. While technology tools may provide warnings, alerts, or temporary assistance (such as data visualization or basic calculations), these do not constitute investment automation as they only support rather than replace human decision-making. All fundamental analysis, valuation work, portfolio construction, and trade execution require direct human involvement.

- **Level 1:** ***Investor/Researcher Assistance.*** AI systems provide sustained assistance with specific subtasks of the investment process but do not integrate multiple functions or provide investment recommendations. Examples include automated data collection from regulatory filings, historical metric retrieval, news aggregation, and earnings date alerts. The AI provides structured information but humans perform all analysis, interpretation, and decision-making. This represents a human-in-the-loop model where AI enhances human productivity without replacing human judgment.

- **Level 2:** ***Partial Investment/Research Automation.*** AI systems perform multiple integrated subtasks by analyzing data from various sources and providing synthesized insights. Capabilities include sentiment analysis of earnings calls, financial ratio trend identification, peer valuation metric calculations, and performance summary generation. The AI highlights areas of interest and provides analytical insights but does not generate actionable investment recommendations. Humans actively supervise all AI outputs, formulate investment theses, and make all buy/sell/hold decisions. The operational design domain is limited to specific analytical tasks within defined parameters.

- **Level 3:** ***Conditional Investment Automation.*** AI systems perform substantial portions of the investment process within well-defined domains. Capabilities include identifying undervalued companies using learned valuation models, constructing preliminary valuations with stated

assumptions, generating draft investment recommendations, and monitoring portfolio companies with adjustments for new information. The AI suggests buy/sell actions subject to human review and approval. Humans set overall strategy, define the AI's operational domain, and must intervene when the system encounters unusual data or events outside its training parameters. This represents a human-on-the-loop model where humans supervise and the AI proposes investment actions.

- **Level 4:** ***High Investment Automation.*** AI systems perform the entire investment task autonomously within defined operational domains, including research, valuation, stock selection, portfolio construction, and trade execution. The system adapts dynamically to evolving fundamentals and market conditions while handling unexpected company news without human assistance. Humans set high-level goals, risk parameters, and ethical guidelines but have no involvement in tactical decisions within the AI's domain. This represents a human-out-of-the-loop model for execution while maintaining human oversight for strategic direction. The system manages all financial events and contingencies within its domain, with human intervention required only for fundamental violations or catastrophic failures.

- **Level 5:** ***Full Autonomous Investing/Research.*** AI systems perform the entire investment process across unlimited operational domains, encompassing all markets, strategies, and conditions that human experts could manage. The system autonomously defines strategies, adapts valuation methodologies globally, synthesizes information from any relevant source, and manages portfolios with creativity and foresight that may exceed skilled human investors. Human involvement is minimal except for setting high-level ethical guidelines or societal objectives. The system detects and responds to any global event affecting investment decisions, including novel situations, and handles all fallback scenarios autonomously. Like Level 5 autonomous driving, this level remains hypothetical and raises important questions about systemic risk and AI alignment in financial markets.

4 Industry Case Studies

The practical application of our L0 to L5 framework becomes clearer through examination of real-world implementations across different levels of autonomy. These case studies demonstrate how AI systems at various levels have been successfully deployed in investment management while highlighting the human oversight and operational boundaries that characterize each level.

4.1 Hedge Fund X and Sleep Apnea Investment Analysis (L1–L2)

Hedge Fund X, a healthcare-focused investment fund, faced a significant challenge when GLP-1 weight-loss drugs emerged as potential disruptors to their portfolio company specializing in sleep apnea devices.

Table 1: Autonomous Driving vs. Autonomous Investing

Level	Autonomous Driving	Autonomous Investing
L0	**No Automation** Driver performs all driving tasks. Warnings only (e.g., blind spot alerts). *Full human control*	**No Investment Automation** All analysis performed manually. Alerts only (e.g., price warnings). *Full human control*
L1	**Driver Assistance** Single function automation (adaptive cruise OR lane keeping). *Human-in-the-loop*	**Investor Assistance** Single function automation (data collection OR metric calculation). *Human-in-the-loop*
L2	**Partial Automation** Multiple functions (cruise AND steering). Constant supervision required. *Human supervises*	**Partial Investment Automation** Integrated analysis (sentiment AND valuation). Human validates outputs. *Human supervises*
L3	**Conditional Automation** System drives in specific conditions. Human responds to takeover requests. *Human-on-the-loop*	**Conditional Investment Automation** AI recommends trades in defined domains. Human reviews and approves. *Human-on-the-loop*
L4	**High Automation** Full self-driving in geo-fenced areas. No human intervention in domain. *Human-out-of-the-loop*	**High Investment Automation** Autonomous investing in defined markets. No intervention in domain. *Human-out-of-the-loop*
L5	**Full Automation** Complete autonomy in all conditions. No steering wheel needed. *Fully autonomous*	**Full Autonomous Investing** Complete autonomy across all markets and conditions. *Fully autonomous*

The stock price of their companies dropped significantly as investors feared that effective weight-loss medications might reduce demand for traditional sleep apnea treatments.

The firm implemented an L1–L2 AI system to address this challenge. At the L1 level, AI tools automated the collection of hundreds of peer-reviewed studies on GLP-1 drugs and their effects on obstructive sleep apnea. The system gathered data from medical journals, clinical trial databases, and regulatory filings, creating a comprehensive dataset that would have required weeks of manual research.

Moving to L2 capabilities, the AI system analyzed the collected research to identify key trends and synthesize findings. The analysis revealed that 42-50% of obstructive sleep apnea patients continued to require airway intervention even after successful weight loss through GLP-1 medications. The AI highlighted studies showing that sleep apnea has multiple contributing factors beyond weight, including anatomical considerations that persist regardless of weight loss.

The human investment team used these AI-generated insights to formulate their investment thesis, determining that GLP-1 drugs represented a complementary rather than replacement therapy for sleep apnea treatment. This analysis enabled the fund to maintain their position in the sleep apnea device manufacturer rather than selling during the market panic, ultimately generating significant returns as the stock recovered once the market better understood the nuanced relationship between weight-loss drugs and sleep apnea treatment.

This case demonstrates the effectiveness of L1–L2 systems in enhancing human analytical capabil-

ities while maintaining clear human responsibility for investment decisions. The AI system operated within a well-defined operational design domain focused on medical literature analysis, with humans retaining full control over interpretation and investment strategy.

4.2 Hedge Fund Y and Decentralized Finance Protocol Evaluation (L2–L3)

Hedge Fund Y, a cryptocurrency-focused investment fund, needed to evaluate over 1,000 decentralized finance (DeFi) protocols for potential investment opportunities. Traditional due diligence approaches would have required 30 minutes to several hours per protocol, making comprehensive coverage impossible within practical time constraints.

The firm developed an L2–L3 AI agent swarm system that automated significant portions of the due diligence process. The system incorporated multiple specialized agents, each focused on specific aspects of protocol evaluation. One agent conducted automated smart contract audits, analyzing code for potential vulnerabilities, inefficiencies, and security risks. Another agent modeled tokenomics, evaluating token distribution, inflation schedules, and incentive mechanisms. A third agent scanned on-chain data to assess protocol usage, liquidity trends, and governance activity.

At the L2 level, these agents provided synthesized insights and risk assessments for each protocol, highlighting key strengths and potential concerns. The system generated standardized evaluation reports that enabled rapid comparison across protocols. Moving toward L3 capabilities, the system began generating preliminary investment recommendations based on predefined criteria established by the human investment team.

The AI system operated within a clearly defined operational design domain that included specific blockchain networks, protocol types, and evaluation criteria. Human oversight remained essential for interpreting results within broader market context and making final investment decisions. However, the system enabled the fund to achieve 95% faster due diligence processes while covering 20 times more protocols than traditional manual approaches.

This enhanced coverage capability allowed Hedge Fund Y to identify investment opportunities significantly earlier than competitors using traditional analysis methods, providing substantial competitive advantage in the rapidly evolving DeFi market. The case demonstrates how L2–L3 systems can dramatically enhance analytical capacity while maintaining appropriate human oversight for complex investment decisions.

4.3 Sovereign Wealth Fund and Manager Oversight (L3)

A sovereign wealth fund managing relationships with 75 external investment managers faced challenges in monitoring portfolio manager performance and risk exposure through traditional quarterly review processes. The existing system provided insufficient early warning capabilities for emerging risks and required extensive manual data compilation that delayed strategic decision-making.

The fund implemented an L3 autonomous investment monitoring system that transformed their

manager oversight capabilities. The system automatically ingested daily performance data, holdings information, and risk metrics from all external managers, creating a comprehensive real-time view of the entire portfolio. Advanced analytics identified patterns and anomalies that might indicate emerging risks or changing manager behavior.

The system generated automated alerts for significant performance deviations, litigation developments, executive turnover, strategy drift, and unusual portfolio concentrations. A unified dashboard provided standardized risk metrics and performance comparisons across all managers, enabling apples-to-apples evaluation despite different investment strategies and asset classes.

The L3 system operated within clearly defined parameters for risk monitoring and alert generation while requiring human oversight for interpretation and action. The operational design domain included specific performance thresholds, risk metrics, and alert criteria established by the fund's investment committee.

The impact of this system proved its worth when it flagged a litigation threat affecting one external manager 47 days before an official disclosure. This early warning enabled the sovereign fund to reduce exposure and avoid significant losses that affected other investors who learned of the risk through traditional disclosure channels.

The system also transformed staff allocation within the fund, enabling investment professionals to shift from manual data compilation and routine monitoring to strategic portfolio allocation and manager selection activities. This represents the value creation potential of L3 systems that handle routine operational tasks while enabling humans to focus on higher-value strategic decisions.

5 Implementation Considerations and Risk Management

We discuss three aspects of successful implementations of autonomous AI in investment management systems: operational design domains, how humans are used in the autonomous AI systems and when humans should intervene, and risk management and systemic risk considerations. We also discuss regulatory and ethical implications of investment autonomous AIs.

5.1 Operational Design Domain Definition

Operational design domains should clearly specify the conditions under which AI systems can operate effectively. These domains must account for market conditions, asset classes, investment strategies, and risk parameters while providing clear boundaries that trigger human intervention when necessary.

Market condition parameters should define the volatility regimes, liquidity environments, and market structure assumptions under which the system is designed to function. For example, an L4 system might operate effectively during normal market conditions but require human oversight during periods of extreme volatility or when market structure changes fundamentally.

Asset class specifications must clearly define which securities, markets, and instruments fall within the system's operational domain. Geographic restrictions, market capitalization limits, and liquidity

requirements provide important boundaries that help ensure system reliability. Currency exposure limits and regulatory jurisdiction constraints represent additional operational boundaries that require careful specification.

Investment strategy parameters should define the philosophical and methodological approaches that the AI system is designed to implement. For example, value investing systems require different operational domains than growth-focused or momentum-based strategies. Risk management parameters, including position sizing limits, concentration restrictions, and stop-loss protocols, provide essential safeguards that prevent system failures from causing catastrophic losses.

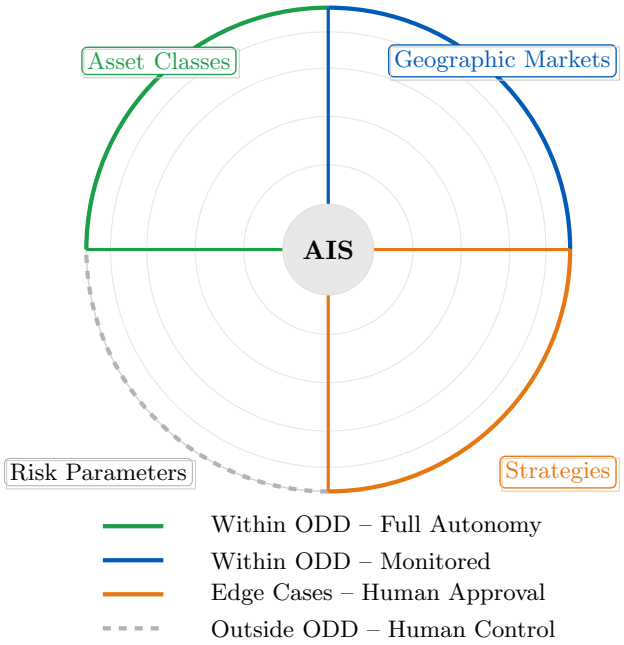

Figure 2: ODD defines where L3–L4 autonomy is permitted; outside requires human fallback.

5.2 Human Oversight and Intervention Protocols

Each level of autonomy requires different approaches to human oversight and intervention, with clear protocols that specify when and how humans should intervene in AI system operations. These protocols must balance the efficiency benefits of automation with the risk management requirements of fiduciary responsibility.

L1 and L2 systems require continuous human oversight with humans maintaining decision-making authority for all investment actions. Intervention protocols should specify quality control procedures for AI-generated analysis, validation requirements for data inputs, and escalation procedures when AI systems produce unexpected or contradictory results.

L3 systems require monitoring protocols that enable humans to supervise AI-generated recommendations without becoming bottlenecks that eliminate automation benefits. Clear criteria should define when AI recommendations require human review versus when they can be implemented automatically. Exception handling procedures should specify human intervention requirements when AI

systems encounter situations outside their training parameters.

L4 systems require strategic oversight protocols that focus on system performance monitoring rather than tactical decision review. Humans should establish performance benchmarks, risk limit monitoring, and periodic system evaluation procedures while avoiding intervention in routine operational decisions within the defined operational domain.

5.3 Risk Management and Systemic Risk Considerations

Implementation of autonomous investment systems creates both firm-specific and systemic risks that require careful management. Firm-level risks include system failures, model drift, operational errors, and cybersecurity vulnerabilities. Systemic risks emerge when multiple firms implement similar AI systems that could exhibit correlated behavior during market stress periods.

- Model risk management becomes particularly important as AI systems become more sophisticated and operate with greater autonomy. Traditional model validation techniques may prove insufficient for complex machine learning systems that adapt continuously to new data. Ongoing monitoring procedures should track model performance, identify drift patterns, and establish criteria for model updates or replacements.

- Operational risk management must address the potential for AI systems to execute trades or make investment decisions faster than human oversight can respond to emerging problems. Circuit breakers, position limits, and automatic shutdown procedures provide essential safeguards against runaway systems. Regular stress testing should evaluate system behavior under various market scenarios and failure modes.

- Systemic risk considerations become increasingly important as autonomous investment systems achieve broader adoption. Regulatory frameworks may need to address the potential for correlated AI behavior to amplify market volatility or create new forms of systemic risk. Industry coordination may become necessary to prevent unintended consequences from widespread AI adoption.

5.4 Regulatory Framework Evolution

The increasing adoption of AI in investment management creates new challenges for regulatory oversight that traditional frameworks may not adequately address. Current regulations focus primarily on human decision-making processes and may require substantial adaptation to accommodate autonomous investment systems.

Fiduciary responsibility regulations must evolve to address situations where AI systems make investment decisions with varying degrees of human oversight. New standards should define acceptable levels of automation for different types of investment activities, while maintaining investor protection

requirements. The allocation of responsibility between AI systems and human supervisors requires careful legal consideration.

Market manipulation and insider trading regulations may need updating to address AI systems that can process vast amounts of information and execute trades at superhuman speeds. Traditional concepts of "material nonpublic information" may require reexamination when AI systems can synthesize publicly available information in ways that provide significant trading advantages.

Algorithmic trading regulations provide a starting point for AI oversight but may prove insufficient for sophisticated investment systems that go beyond simple trade execution. New frameworks should address AI system transparency, auditability, and accountability while avoiding regulatory approaches that stifle beneficial innovation.

5.5 Ethical Considerations and AI Alignment

Finally, the deployment of autonomous investment systems raises important ethical questions about the goals and constraints that should guide AI behavior in financial markets. These systems will wield significant influence over capital allocation decisions that affect companies, investors, and broader economic outcomes.

AI alignment challenges emerge when autonomous systems optimize for metrics that may not fully capture human values or long-term interests. For example, investment AI systems trained solely on historical return maximization might develop strategies that produce short-term gains while creating long-term market instability or exacerbating wealth inequality.

Transparency and explainability requirements become particularly important for autonomous investment systems that make decisions affecting other people's money. Investors and regulators need sufficient understanding of AI decision-making processes to evaluate system reliability and appropriateness. The black box nature of many AI systems may prove incompatible with fiduciary responsibility requirements.

Fairness and bias considerations require attention to ensure that AI investment systems do not perpetuate or amplify existing market biases or create new forms of discrimination. Investment decisions based on biased training data or flawed algorithmic assumptions could systematically disadvantage certain companies, sectors, or regions in ways that harm economic efficiency and fairness.

6 Future Research Directions and Industry Evolution

As we look ahead to the future of AI in investments, we highlight three considerations: the future capabilities of AI, how it will affect the structure of the investment industry, and our current knowledge gaps that need to be addressed if autonomous investment systems are to perform reliably, transparently, and at scale.

6.1 Technological Development Pathways

The evolution toward higher levels of investment automation will likely follow technological development pathways that address current limitations in AI capabilities. Natural language processing advances will enhance AI systems' ability to interpret complex financial documents, management communications, and market commentary. Computer vision applications will soon make available to everyone analysis of satellite imagery, facility photos, and other visual information relevant to investment decisions.

Multimodal AI systems that integrate text, audio, video, and numerical data will provide more comprehensive analytical capabilities than current systems that focus primarily on structured financial data. These systems may identify investment opportunities by synthesizing information across previously disconnected data sources.

Reinforcement learning applications in investment management remain largely unexplored but offer potential for AI systems that learn from market feedback and adapt their strategies continuously. These systems could potentially develop novel investment approaches that exceed human performance in specific domains.

Large language models specialized for financial applications may enable more sophisticated analysis of qualitative factors that traditionally require human judgment. These systems could analyze management quality, competitive positioning, and industry dynamics with greater consistency and comprehensiveness than human analysts.

6.2 Industry Structure Evolution

The widespread adoption of autonomous investment systems will likely transform the industry landscape in ways that create both opportunities and challenges for different types of investment firms. Large institutional investors with extensive technology resources may gain significant advantages in developing and deploying advanced AI systems, potentially creating competitive pressures that disadvantage smaller firms.

Technology companies may increasingly compete directly with traditional investment managers by offering AI-driven investment services. This convergence of technology and finance could blur industry boundaries and create new competitive dynamics that challenge existing business models.

Investment management may evolve toward greater specialization, with some firms focusing on developing AI technology, while others focus on human oversight, strategy development, or specialized market segments that benefit less from automation. This specialization could create new forms of collaboration and partnership within the industry.

The democratization of sophisticated investment tools through AI could enable smaller investors to access analytical capabilities previously available only to large institutions. This trend might reduce some historical advantages of scale while creating new opportunities for innovative investment approaches.

6.3 Research Priorities and Knowledge Gaps

Several important research questions require attention as the industry moves toward greater investment automation. The optimal allocation of responsibilities between humans and AI systems under different market conditions and investment strategies remains poorly understood. Research should examine which tasks benefit the most from automation versus human judgment.

There are systemic risk implications of the widespread adoption of AI in investment management. Research should evaluate the potential for correlated AI behavior to increase market volatility and develop frameworks to manage these risks. Agent-based modeling and simulation studies could provide insight into system-level effects that are difficult to observe in real markets.

Performance evaluation methodologies for AI investment systems need to be developed to address the unique characteristics of machine learning approaches. Traditional performance attribution techniques may prove insufficient for AI systems that continuously adapt and make decisions based on complex, nonlinear relationships in data.

The effects of interaction between multiple AI systems operating in the same markets represent an important area of research. As AI adoption increases, understanding how these systems influence each other and aggregate market behavior becomes crucial to maintaining market stability and efficiency.

7 Conclusion

We present a framework for classifying AI autonomy in investment management, adapting the proven success of the SAE J3016 standard for autonomous driving for the investment management industry. The L0–L5 taxonomy provides clear delineations of system capabilities, human responsibilities, and operational boundaries that enable more effective communication, development, and oversight of AI-driven investment systems. Three case studies demonstrate practical application of AI in investing across different levels of autonomy: progressing from AI assistance (L1-L2) through conditional automation toward the higher levels of autonomy (L3) that represent the future direction of the industry.

The framework addresses several pressing challenges facing the investment management industry as it navigates the transformative potential of AI. Clear definitions of the domain of operational design, robust human oversight protocols, and comprehensive risk management frameworks are essential for successful AI integration. By establishing standardized terminology and clear expectations for AI capabilities, the framework reduces ambiguity and enables more informed decision making about the adoption and integration of technology. The explicit definition of human oversight responsibilities and operational design domains provides guidance for risk management and regulatory compliance.

Looking ahead, the evolution toward higher levels of investment automation appears inevitable as AI technologies continue to advance and competitive pressures drive adoption. The L0–L5 framework provides a foundation for this evolution by establishing clear benchmarks for progress and common standards for evaluation. With such a structured approach to understanding and implementing autonomous investment systems, AI can be used in such a way that it benefits investors, strengthen mar-

kets, and supports broad economic prosperity. However, realizing the full benefits of autonomous investment systems will require continued collaboration among technology developers, investment firms, regulators, and academic researchers.